Noise and its limiting effect on communication.

There are several sources of noise in communications systems. The most limiting type is thermal noise, caused by the random motion of electrons in conductors primarily. These motions cause bodies to radiate weak amounts of electromagnetic energy, which is proportional to the Absolute temperature T. Any body in thermal equilibrium with its surroundings absorbs from other bodies a power equal to that radiated by itself, so there is no net loss or gain of energy.

In resistive circuits, the mean of the noise voltage squared which is generated can be shown by thermodynamics to be proportional to the resistance R :-

$$e_n^{\,2} \;\alpha\; R$$

Also, the noise voltage squared, $e_n^{\,2}$, measured across **R** is proportional to the bandwidth **B**, wherever the bandwidth of interest is situated. Clearly, minimisation of the bandwidth for a given application preserves the Signal to Noise ration (SNR).

Because noise is a random variable with time, it is usual to refer to its statistics and its ***root mean square*** (RMS) value. In effect noise provides a random amplitude variation around a mean value which forms the average value of the signal. The mean square value is given by :-

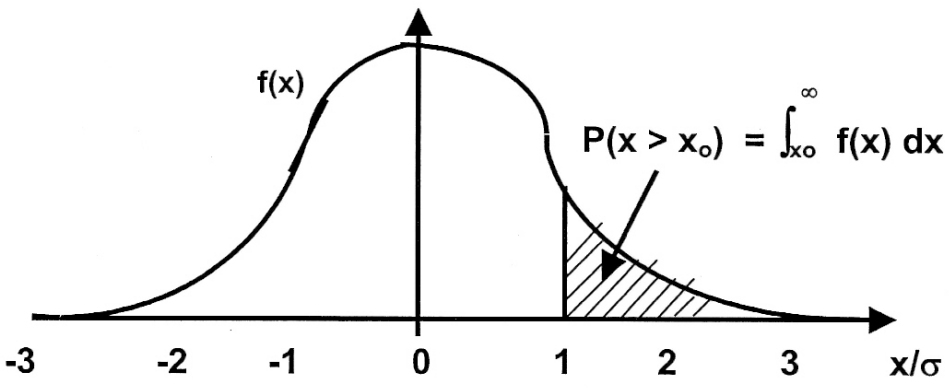
$$\overline{x^2(t)} = \underset{\text{as } T \to \infty}{\text{Limit}} \frac{1}{T} \int_{-T/2}^{+T/2} x^2(t)\, dt$$

In evaluation of communications systems, it is often necessary to know about the probability of a given amplitude of signal being exceeded, due to noise fluctuations, especially digital systems in which a sharply-defined threshold is set for data detection.

It may be shown that the majority of noise signals follow a Gaussian distribution.

$$f(x) = \frac{\exp(-x^2/2\sigma^2)}{\sigma(2\pi)^{1/2}} \quad \text{where} \quad \sigma^2 = \overline{x^2}$$

σ is the standard deviation, which, in this case, is equal to the RMS value of the waveform. The distribution has the form below :-



$P(x > x_o)$ means the probability that x exceeds x_o , found by evaluating the integral. The total area under the curve : -

$$\int_{-\infty}^{\infty} f(x)\, dx = 1$$ i.e. the total of all probabilities is unity.

Shot noise

Suppose an electronic device has a current flowing through it, which consists of a series of random impulses of current due to a flow of electrons. Let the current be **i(t)**. Let the time taken for an electron to cross the device be τ . i(t) is zero if t is less than zero, and zero when $t > \tau$. Therefore :-

$$\int_0^{\tau} i(t)\, dt = e = \text{charge on electron}$$

If the number of electrons emitted per second is **N** , the total charge transferred per second is **Ne** , equal to the device current I_o .

$$I_o = Ne$$

As the electrons are emitted at random instants, we may use the statistical results that the mean square values can be added. It is justified that the transit time τ is negligibly short. This means that **i(t)** can be represented by an impulse function such that the following is derived from the expressions above:-

$$i(t) = e.\ \delta(t)$$

The frequency spectrum of a single pulse $i(t - t_n)$ which occurs at time t_n is given by :

$$g_n(\omega) = e.\ \exp(-j\omega t_n)$$

giving :

$$|g_n(\omega)|^2 = e^2$$

The magnitude of $g_n(\omega)$ is independent of the time at which the pulse occurs.

The value of $|\, g_n(\omega)\, |^2$ for the NT pulses which occur in a period T is given by adding the values of $|\, g_n(\omega)\, |^2$ for the individual pulses, and the power spectrum of $G(\omega)$ is thus :-

$$G(\omega) = \lim_{T \to \infty} \frac{NT\,|\, g_n(\omega)\, |^2}{T}$$

$$= e.I_o$$

This power spectrum is the value of the mean square current associated with a bandwidth of 1 Hertz, centred on angular frequency ω . The spectrum is defined, at least mathematically, by positive- and negative frequencies, and there are, therefore, equal contributions from the bands, centred on $+\omega$ and $-\omega$. If the bandwidth is **df** for each band, then the resulting noise is : -

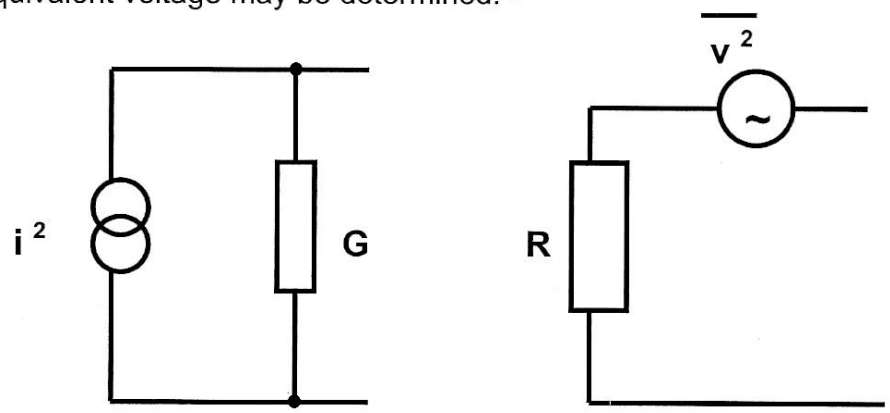
$$\overline{i^2} = 2\, e.I_o\, df \quad \text{(Schottky formula)}$$

Thermal noise.

Equivalent analysis may be undertaken to determine the mean square noise current or voltage associated with a resistor at an Absolute temperature T :-

$$\overline{i^2} = 4kTG\, df \quad \text{(Johnson noise formula)}$$

where **k** = Botzmann's constant (1.37×10^{-23} Joule/degree Kelvin), G is the conductance of the resistor, and **df** is the bandwidth of measurement, similarly to the shot noise situation. By a simple transformation, the noise equivalent voltage may be determined: -



$$\overline{v^2} = \overline{[R\,i]^2} = R^2 . \overline{i^2}$$

By Thevenin's theorem, therefore : -

$$\overline{v^2} = 4kTR\,df$$

As it may be seen that the noise power (proportional to $\overline{v}^2$ and $\overline{i^2}$) is proportional to bandwidth, this noise is defined as ***white***. The above formulae apply to resistors, and also aerials (which, electrically, can be regarded as having resistive components). Note that the noise power for all resistances is given by the expression : -

$$P_n = 4kT\,df$$

If a resistance R1 is connected across another resistance R, noise power will be transferred from R to R1. If R1 is at the same temperature as R, an equal amount of noise power must be transferred from R1 to R, otherwise there is a violation of thermodynamics laws. Maximum power transfer from R to R1 occurs if R1 is equal to R (matched load), and the power transferred is : -

Voltage across **R1** = voltage across **R** = $\overline{v}/2 = \overline{v_R}$

Therefore, power transferred = $\overline{v_R}^2 / R = kT\,df$

This is called the ***available noise power***, and it is independent of the value of R.

Signal to Noise Ratios

The Signal to Noise Ratio for a system is usually defined by the following :-

$$SNR = 10 \log_{10} \left(\frac{S}{N} \right)$$

where **S** is the signal power, and **N** is the average noise power.

In most cases, the signal to noise ratio (SNR) can be measured as the mean of the (signal + noise) power, when $\mathbf{S >> N}$, usually for convenience of measurement. The noise power is normally measured when the signal is turned off, except in circumstances where the noise is a function of signal level (e.g. shot noise-dominated system).

It is now useful to define ***Noise Figure*** for a communication channel. Consider a channel with an input signal power $\mathbf{S_1}$ and a power gain **G**. The output power is thus $\mathbf{G.S_1}$. The input noise power, N_1 is also amplified and contributes an output noise power $\mathbf{G.N_1}$. In addition, noise is introduced within the channel and so the total noise output, $\mathbf{N_2}$, will exceed $\mathbf{G.N_1}$.

The SNR at the output $\mathbf{G.S_1/N_2}$ is therefore less than that at the input $\mathbf{S_1/ N_1}$, since $\mathbf{N_2}$ exceeds $\mathbf{G.N_1}$. A measure of the degradation caused by the channel in terms of SNR is : -

$$\text{Noise factor} = F = \frac{\text{SNR at input}}{\text{SNR at output}}$$

$$= \frac{S_1/ N_1}{G.S_1/N_2} = \frac{N_2}{G.N_1}$$

An ideal channel introduces no extra noise, so $\mathbf{F = 1}$. The noise factor is often expressed in dB (i.e. as $\mathbf{10 \log_{10} F}$) as the output SNR in dB is then found from : -

$$SNR_o = SNR_i - F_{(dB)}$$

The noise factor of a communication channel can be determined by referring all noise back to the input of an equivalent, noise-free amplifier. Using this approach, the noise figure of a number of cascaded stages can be determined as : -

$$F = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1.G_2} + \frac{F_4 - 1}{G_1.G_2.G_3} \ldots\ldots$$

It is very important, therefore, to have the first stage gain very large and for this stage to have a low noise figure, in order for the whole chain to have an overall low noise figure.

Another approach is possible in systems utilising an aerial, for example, of *radiation resistance*, **R**. If the aerial is matched to the receiver, $\mathbf{N_1}$ is the available noise power of the radiation resistance and has the value $\mathbf{k.T_1.B}$ where $\mathbf{T_1}$ is the effective temperature of the aerial and B is the receiver bandwidth in Hz. In this case, $\mathbf{N_1}$ is not a constant and **F** will vary as $\mathbf{N_1}$ varies.

The input noise to the amplifier is given by $\mathbf{k.T_e.B}$, where $\mathbf{T_e}$ is defined as the effective noise input temperature of the amplifier. $\mathbf{T_e}$ can be regarded as the increase in temperature of the aerial to account for noise generated in the amplifier. $\mathbf{T_e}$ is a constant for the receiver, independent of the effective temperature of the aerial.

If the temperature of the aerial is $\mathbf{T_1}$ and the effective noise temperature of the receiver is $\mathbf{T_e}$, the total noise input power is $\mathbf{k.(T_1 + T_e).B.}$ A noiseless receiver has an input noise power $\mathbf{k.T_e.B}$, so that the noise factor may be derived as : -

$$F = \frac{(T_1 + T_e)}{T_1} = 1 + T_e / T_1$$

In many situations, the noise temperature of the input circuit is room temperature, typically taken as 290° K . If $\mathbf{T_e}$ is 60° K , then : -

$$F = 1 + 60/290 = 1.21 \text{ , or } 0.8 \text{ dB}$$